

POKHARA UNIVERSITY

Level: Bachelor Semester: Fall Year : 2023
 Programme: BE Full Marks: 100
 Course: Database Management System (New) Pass Marks: 45
 Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is data abstraction in DBMS? Explain in detail. 7
- b) Consider you are asked to design a database for the Exam section of your college. Draw its ER diagram assuming required entities and their attributes. 8
2. a) Convert the ER diagram that you designed in question no 1 b) into relational schema. 7
- b) What are the views? Consider the table **tbl_emp** as follows: 8

Emp_id	Emp_name	Salary	Department	Date of joining
101	Anish	20000	Packing	2070-02-01
102	Rojina	18000	Cleaning	2075-04-06
103	Sita	35000	Polishing	2078-09-12

Write the SQL statements for the following:

- i. Create the above table by considering Emp_id as primary key and insert the above records.
- ii. Change the Department of Anish to marketing.
- iii. Increase the salary of employees whose department is Cleaning by 12%.
- iv. Find the name of employees having salary greater than 16000 and who joined after 2072-11-25
- v. Add a new column Address to the above table.
- vi. Delete the entire table.

OR

Why is joining in SQL necessary? Explain Inner Join, Natural Join and Outer Join with suitable examples.

3. a) What are the different types of integrity constraints? Explain with examples. 8

- b) What is denormalization? Why is it necessary? Explain in detail. 7
4. a) What is the role of views in security? Explain the concept of authorization with suitable examples. 7
- b) Explain the basic steps in query processing in detail. 8
5. a) Explain how the records are organized using fixed-length and variable-length records. 7
- b) What is serializability and why is it needed? Explain the ACID properties in brief. 8
6. a) What is crash recovery? Explain log-based recovery method with example. 7

OR

What is transaction rollback? Explain how the Remote Backup System provides high availability and recovery facility.

- b) List out the different categories of NoSQL databases. Explain the concept of blockchain with its properties. 8
7. Write short notes on: (Any two) 2×5
- a) Nested Queries
- b) Third Normal Form (3NF)
- c) Lock-based protocols for concurrency control

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Attempt all the questions.

1. a) What is a database management system? Difference between Physical and Logical Data Independence. 7
- b) Draw E-R diagram for airline reservation system. The system must keep track of customers and their reservations, flights and their status, seat assignments on individual flights. Include appropriate relationship and cardinality. 8

OR

2. a) Draw an ER diagram of an "Exam System of Pokhara University" using extended features. Explain Primary key and Foreign key. Consider the following schema: 7
 STUDENT(Student_ID, Student_name, Major)
 COURSE(Course_ID, Course_name, Credits)
 ENROLLS_IN(Student_ID, Course_ID, Grade)
 Write the relational algebra expressions for the following cases:
 i. Find the names of all students who are enrolled in courses with names that start with "C".
 ii. Update the grades of students to 'A' whose current grade is 'B'.
 iii. Find the average credits of courses each student is enrolled in.
 iv. Update the name of the course "Physics" to "Advanced Physics".

- b) Let us consider the following relation: 8
 Sailors (sid, sname, rating, age)
 Boats (bid, bname, color)
 Reserves (sid, bid, day)
 Write a SQL statements for the following:
 i. Find the records of sailors who have reserved boat number 75 (bid=75).

- ii. Update the color of the yellow boat to green.
- iii. Delete the records of sailors whose rating is less than 5.
- iv. Find the name and rating of sailors who have reserved a black or blue boat.
- v. Find the age of sailors who have reserved boat number 25 on day 3.

3. a) Define integrity constraints? Explain different integrity constraints with suitable examples. 8

OR

- Why is database normalized? Explain about 1NF, 2NF, 3NF & BCNF.
- b) Explain different types of functional dependencies with examples. Find the closure of attributes A, AC and ABC for the relation R(A,B,C,D) where $A \rightarrow B, BC \rightarrow D$ are the dependencies in R. 7
4. a) Compare authentication and authorization. Explain the different types of access control mechanisms. 7
- b) How query optimization is carried out? Explain about cost estimation of query. 8
5. a) Construct a B+ tree of order 4 for the given key values assuming empty tree initially and keys arranged in ascending order: {4,8,1,20,5,15,16,7,9,38,34,6,39,25} 7
- b) What is serializability? Explain how concurrency control protocols are used? 8
6. a) What are NoSQL databases? Explain the advantages of NoSQL databases. 7
- b) Differentiate between centralized and distributed databases. Explain in brief about NOSQL. 8
7. Write short notes on: (Any two) 2×5
 a) Equivalence of expressions
 b) Stored Procedures
 c) ACID Property

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Attempt all the questions.

1. a) Differentiate between schema and instances. Explain data independence with its importance. 8
 b) Design an ER diagram for an Online Food Delivery System to manage customers, restaurants, orders, and delivery. Draw the ER diagram with cardinalities and constraints. 7
2. a) What is relational algebra? Explain Selection, Projection, Union, Cartesian-product, Join and Set-Difference operations with suitable examples. 7
 b) Consider the given relational schema: 8
 Movie(Title, Director, Myear, Rating)
 Actors(Actor, Age)
 Acts(Actor, Title)
 Directors(Director, DirAge)
 Write the SQL statement for the following task:
 i. Find the movies made by Gopal Varma after 1997.
 ii. Find all actors and directors.
 iii. Find number of movie which are directed by Devid Dhaban having actor "Bishnu".
 iv. Find (Director, Actor) pairs where the directors is younger than the actor.
 v. Find the number of movie actor wise.
3. a) Explain functional dependency with its types? For $R = \{A, B, C, D\}$ and $F = \{A \rightarrow BC, B \rightarrow C, AB \rightarrow C, AC \rightarrow D\}$, List the closure of functional dependency F. 7

OR

What are database constraints? Explain its types with suitable examples. When do we need Denormalization? Explain.

- b) Define normalization. Create a relation in Unnormalized Form (UNF) and normalize it up to the Second Normal Form (2NF). 8
4. a) Why security is needed in database? Explain access control mechanisms. 7

OR

As a database administrator, purpose the measures to protect the database against physical and cyber threats.

- b) Define ACID properties. Explain view serializability with suitable examples. 8
5. a) What do you mean by query processing? Explain the steps of query processing with necessary diagram. 7
 b) Construct a B+ tree for the following set of key values: $\{2,5,7,9,11,17,21,23,29,31\}$. Assume that the tree is initially empty and values are added in ascending order where the pointer number is FOUR. 8
6. a) Consider following two transactions T1 and T2 given in figure where T1 executes before T2. Also consider initial value of A, B and C are 1000, 2000 and 3000 respectively. 7

T1	T2
Read (A)	Read (C)
$A = A - 500$	$C = C - 200$
Write (A)	Write (C)
Read (B)	
$B = B + 500$	
Write (B)	

Use the concept of deferred and immediate database modification log based recovery method in given problem.

- b) Define data warehousing and explain how it supports decision-making processes in large organizations. 8
7. Write short notes on: (Any two) 2×5
 a) Query cost estimation
 b) Set operations in SQL
 c) NoSQL Databases and its characteristics

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Attempt all the questions.

1. a) Describe the architecture of a Database Management System and Database Application (1-tier vs 2-tier vs 3-tier). 7
- b) Create an ER diagram for a movie rental store where: 8
 - i. Customers rent movies.
 - ii. Movies have movie ID, title, genre, and release year.
 - iii. Each customer has a customer ID, name, and membership type.
 - iv. Capture the rental date and return date for each movie rented.
2. a) Consider the following relational database where the primary keys are underlined: 8

Empdetail (emp-name, street, city)
 Jobdetail (emp-name, university-name, salary)
 Universitydetail (university-name, city)
 Headdetail (emp-name, head-name)

Express the following queries in relational algebra:

 - i. Give all employees of Pokhara University a 15 % salary raise.
 - ii. Update the city of Amit to Butwal.
 - iii. Give all managers in this database a 25 percent salary raise.
 - iv. Find all employee name who work for Pokhara University and has salary greater than 70000.
- b) Consider the relational database with the following relations: 7

Student(StudentID, Name, DeptID, Age)
 Department(DeptID, DeptName, HOD)

Enrollment(EnrollID, StudentID, CourseName, Grade)

Now write the SQL command for the following:

- i. Write SQL statements to create the student table.
 - ii. Display each student's name, department name, and course name for all enrolled students.
 - iii. Display the average grade for each course, based on the Enrollment table and group the result by CourseName.
 - iv. Update the age of student 'Iris' to 26.
3. a) What is Domain, Entity Integrity, and Referential Integrity constraints? Explain with examples. 8

OR

- Define normalization and explain its purpose in database design. Discuss the problems (anomalies) that occur in unnormalized databases.
- b) What is relational model? What are the features of good relational database design? Explain. 7
 4. a) Explain access control, authorization, and views in database security. 7
 - b) What is an execution plan? Describe how equivalence rules are used in optimizing queries in DBMS. 8
 5. a) Define indexing. Differentiate between B+ Tree Indexing and Hash Indexing. 7

OR

- Discuss the organization of records in Heap and Indexed Sequential file systems.
- b) What is Serializability? Explain the lock-based concurrency control. 8
 6. a) What is a transaction log? How are undo and redo operations handled using log records? Explain. 8
 - b) Discuss the advantages and limitations of NoSQL databases. Explain different categories of NoSQL. 7
 7. Write short notes on: (Any two) 2×5
 - a) Data Independence
 - b) DDL and DML
 - c) Denormalization

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Attempt all the questions.

1. a) Explain the three levels of data abstraction in DBMS with suitable examples. Also define data independence and why it is important. 7
- b) Design an ER Diagram for a "Hospital Management System." Your diagram must include: 8
 - i. At least four entity sets (e.g., Patient, Doctor, Room, Medicine).
 - ii. Examples of multi-valued and derived attributes.
 - iii. Clearly marked participation constraints and cardinalities.
2. a) Given the relations: 7

Student(SID, Name, Major)
 Course(CID, CName, Credits)
 Enroll(SID, CID, Grade)

Write relational algebra expressions for the following:

 - i. Find the names of students who have a grade of 'A'.
 - ii. List all SIDs of students who have not enrolled in any course.
 - iii. Perform a Natural Join between the two tables.
 - iv. Find the names of students enrolled in 'CS101'.
- b) Consider the relational database 8

Employees (EmpID, Name, DeptID, Salary, JoinDate)
 Departments (DeptID, DeptName, Location)
 Projects (ProjectID, ProjectName, Budget, DeptID)

Write SQL queries for:

 - i. Create the table Employees with appropriate constraints.
 - ii. Find the Name, Salary of employee of 'IT' department.

- iii. Increase the salary of employees in the "IT" department by 10%.
 - iv. Delete all projects with a budget less than 40,000.
3. a) Explain about different types of constraints in Database? Explain each with suitable example. 7
 - b) Explain about different anomalies we face before normalization? Also explain about 2NF, 3NF, BCNF with suitable example. 8

OR

- What is closure of attribute sets? How is it computed? For $R = (A, B, C, D, E, F)$ and $F = \{A \rightarrow BC, E \rightarrow CF, B \rightarrow E, CD \rightarrow EF\}$, Compute $(AB)^+$.
4. a) What is access control in a database system? Explain different access control mechanisms used in DBMS. 7
 - b) What is Query Processing? Explain the steps involved in query processing with suitable diagram. 8
 5. a) Describe different file organization techniques used in DBMS. Explain heap file organization with its merits and demerits. 7
 - b) What is Conflict Serializability? Explain how the Two-Phase Locking (2PL) protocol ensures that a schedule is serializable. 8

OR

- Explain Lock Based Protocol along with its pitfalls. Let we have two accounts A and B with values 5000 and 10000 respectively. Transaction T1 transfers 1000 from A to B. Transaction T2 display sum of A and B. Apply appropriate Locks in above transaction T1 and T2 and make serial schedule.
6. a) Describe the Log-Based Recovery mechanism. Differentiate between Deferred Database Modification and Immediate Database Modification techniques. 7
 - b) What is Object Relational Mapping (ORM)? Explain its role in modern application development. 8
 7. Write short notes on: (Any two) 2×5
 - a) NoSQL Databases and its characteristics
 - b) Transaction State Diagram
 - c) Stored Procedure